

Rotating Rails' PostgreSQL password

Follow the following steps in order to rotate the password of the PostgreSQL user used by the Rails application.

The process involves creating temporary credentials to be used by the application while we rotate the old password, in order to avoid running into unauthenticated connection errors.

Create a CI change request, add the steps below to the request, changing the following:

- [old-username] to the current username used by the Rails application, usually it is gitlab
- [temp-username] to anything reasonable, e.g. gitlab-app
- <env> to the target environment identifier, e.g. gstg, gprd, dr, ...

There are inline comments, to explain some of the steps, pay attention to them.

1. [] On a database node, run ``sudo -u gitlab-psql /usr/lib/postgresql/9.6/bin/pg_dumpall ->`
`<!--`
If the old username contains a dash, then you need to wrap the username in double-quotes

An output will be like:
`CREATE ROLE gitlab;`
`ALTER ROLE gitlab WITH NOSUPERUSER INHERIT NOCREATEROLE NOCREATEDB LOGIN NOREPLICATION M`
`ALTER ROLE gitlab SET statement_timeout TO '10s';`
`-->`
1. [] Copy the output, replace all occurrence of ``[old-username]`` with ``"[temp-username]"``
1. [] In the output, the ``ALTER ROLE ... WITH ...`` line, replace ``PASSWORD`` with ``ENCRYPTED`
1. [] On the primary DB, in a psql console, run updated output
1. [] On the primary DB, in a psql console, run ``GRANT [old-username] to "[temp-username]"``
1. [] In ``gitlab-omnibus-secrets`` and ``gitlab-omnibus-secrets-geo`` GKMS vaults, under ``omni`
 1. [] Update ``db_password`` to ``[temp-username]`'s password`
 1. [] Update ``db_username`` to ``[temp-username]```<!--`
Ideally, ``db_username`` is updated in a Chef role, but since there would a delay between
and updating the vault, there's a chance of clients getting half of the change (i.e. ``[`
``[temp-username]`'s password or `[temp-username]` with `[old-username]`'s password).

Adding both db_username and db_password to the vault is important to have the new creden
-->`
1. [] Update ``omnibus-gitlab.gitlab_rb.gitlab-rails.db_username`` in ``<env>-base`` Chef role
`<!--`
Example: https://ops.gitlab.net/gitlab-cookbooks/chef-repo/-/merge_requests/3157/diffs
`-->`
1. [] Converge Chef on one client node and check for errors
 1. [] If the single node fails to connect to the database, revert the change in GKMS as

1. [] Converge Chef on the client nodes or wait for 30 minutes
1. [] Follow these [instructions](https://gitlab.com/gitlab-com/runbooks/-/blob/e59f5321ae to reflect the new changes in K8s.
1. [] Make sure no client is using or trying to use `[old-username]` by running all of the
 1. [] `knife ssh 'roles:<env>-base -roles:<env>-base-bastion' 'sudo test -f /var/opt/g
 - * Should be no output or output from hosts that has `reconfigure` disabled, the latt
 1. [] `knife ssh roles:<env>-base-db-patroni "sudo gitlab-psql -c \"SELECT COUNT(*) FR
 - * All counts should be zero
 1. [] `knife ssh 'roles:<env>-base-db-patroni roles:<env>-base-db-pgbouncer-pool' 'sudo
 - * No recent entries should exist
1. [] On the primary DB, in a psql console, change the password for the old username by run
1. [] Update the password in 1Password or create a new entry for it. Entry name should be `
1. [] In `gitlab-omnibus-secrets` and `gitlab-omnibus-secrets-geo` GKMS vaults, under `omni
 1. [] Update `db\_password` to `[new-password]`
 1. [] Update `db\_username` to `[old-username]`
1. [] Update `omnibus-gitlab.gitlab\_rb.gitlab-rails.db\_username` in `<<env>-base` Chef role


```
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 1. [] Check Sentry for any errors related to failed DB login attempts
1. [] On the primary DB, in a psql console, drop `[temp-username]` by running `DROP ROLE `
1. [] In `gitlab-omnibus-secrets` and `gitlab-omnibus-secrets-geo` GKMS vaults, remove `omni